Machine Learning - Data Clustering Project

Project Link:

1. Github Link https://github.com/karan-dev-ctrl/Machine-Learning---Data-Clustering-project

Performed Activities :

1. Dataset: Google Review Ratings
2. Techniques Used: KMeans, DBSCAN, Spectral Clustering
3. Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib

Dataset Description:

• Total Records : 5456 users
• Total Features : 24 average review rating categories
• Rating Range : 0 to 5 (Continuous Numerical Data)
• Data Type : Unsupervised clustering dataset

Data Preprocessing:

• Removed User ID and unwanted columns
• Converted features to numeric format
• Filled missing values using median imputation
• Scaled the dataset using StandardScaler
• Applied PCA for 2D cluster visualization

Experiments performed and its comparison:

Method	**K-Means**	**DBSCAN**	**Spectral Clustering**
Type	Partition-Based	Density-Based	Graph-Based
Silhouette Score	0.145	0.669	0.099
Noise Handling	No	Yes	No
Best Overall	Moderate	Best Results	Not Good

Conclusion:

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Based on silhouette evaluation and cluster separation results, **DBSCAN performed the best with a silhouette score of 0.669**. This indicates highly distinct user behavior clusters compared to K-Means and Spectral Clustering, which formed weaker cluster boundaries. The dataset naturally fits density-based clustering, making DBSCAN the most suitable model for this dataset.